Amortized complexity of smoothness-indexed penalized least squares

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ABSTRACT

A normalized trace-based smoothness index provides an interpretable coordinate for finite-difference penalized least-squares trend estimation, but its numerical cost depends on how smoothing parameters, fitted series and statistical selection criteria are reused. We formalize complete criterion-evaluation work as a function of sample length, difference order, number of candidate smoothness values, number of series and requested criterion set. For fixed reference-difference means, we establish the data-independent smoothness-to-penalty mapping and show that banded Cholesky with selected inversion is sufficient to evaluate conditional cross-validation, generalized cross-validation, information criteria and a profiled Gaussian marginal likelihood without a dense inverse. A log-determinant identity supplies a banded expression for the reduced-coordinate likelihood. We derive arithmetic and storage bounds for reusable spectral and streamed banded pipelines, identify an explicit order-one eigenvalue simplification and characterize when a calibrated candidate-count crossover may exist. The results distinguish theoretical operation counts from execution time, account for numerical admissibility and explain why no solver is universally optimal. The analysis synthesizes established smoother, factorization and selected-inversion theory rather than proposing a new smoothing penalty. Two supporting illustrations distinguish measured one-worker BIC pipeline timings from the theoretical bounds and relate a third-difference smoother to a real vehicle-position segment. Broader runtime and forecasting evaluations remain separate research studies.

1. Introduction and smoothness coordinate

Quadratic finite-difference penalized least squares (PLS) has a long history in Whittaker–Henderson smoothing (Whittaker, 1923; Guerrero, 2007). The penalty λ is mathematically convenient, but the same numerical value need not express the same smoothness for different difference orders or sample lengths. Guerrero's trace-based smoothing measure (Guerrero, 2008) and its use with CV, GCV, AICc and BIC (Cortés-Toto, Guerrero and Reyes, 2017) are established precedents. We do not claim to invent the smoother or its tuning criteria; instead, we investigate the *computational cost of an interpretable bounded coordinate and the reuse of its mapping across many evaluations.*

For $y \in \mathbb{R}^n$, difference order $1 \le d < n$, and fixed mean of the penalized differences μ, let $D_d \in \mathbb{R}^{(n-d)\times n}$ denote forward differences and define

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau}\{\|y-\tau\|_2^2 + \lambda\|D_d\tau - \mu\mathbf{1}\|_2^2\}. \quad (1)$$

Put $r = n-d$, $Q_d = D_d^\top D_d$, and $H_\lambda = (I+\lambda Q_d)^{-1}$. A polynomial particular solution is $p_{\mu,t} = \mu\binom{t-1}{d}$, and $\widehat{\tau} = p_\mu + H_\lambda(y-p_\mu)$. The penalty has a d-dimensional polynomial nullspace, so $\mathrm{tr}(H_\lambda) \to d$ as $\lambda \to \infty$.

Write $\delta_1, \dots, \delta_r > 0$ for the positive eigenvalues of Q_d. We use the normalized smoothness index

$$S_d(\lambda) = \frac{n - \mathrm{tr}(H_\lambda)}{n-d} = \frac{1}{r}\sum_{j=1}^{r}\frac{\lambda\delta_j}{1+\lambda\delta_j}. \quad (2)$$

Each $\delta_j > 0$ implies $S_d'(\lambda) = r^{-1}\sum_j \delta_j/(1+\lambda\delta_j)^2 > 0$, hence S_d is continuous, strictly increasing, and maps $[0,\infty]$ onto $[0,1]$. Every interior s has a unique penalty $\Lambda_{n,d}(s)$. The map depends only on (n,d), *not* on observations or the fixed reference mean μ. Therefore it can be cached across different series, criteria and μ values. The denominator $n-d$ normalizes an established trace-based index; it is not a new statistical model.

1.1. Shift-free full spectral reconstruction

Let $D_d = U\,\mathrm{diag}(\sigma)V^\top$ be a reduced SVD over the r positive singular modes and write $w_j = \lambda\sigma_j^2/(1+\lambda\sigma_j^2)$. A numerically useful identity is

$$\widehat{\tau}_{\lambda,\mu} = y - V\big[w \odot (V^\top y) - \mu(w \oslash \sigma) \odot (U^\top\mathbf{1})\big], \quad (3)$$

where multiplication and division are componentwise. It avoids forming the potentially enormous polynomial p_μ at high d and costs $O(nr)$ for a complete curve once the SVD is available. A separate high-precision companion manuscript evaluates its numerical accuracy. Here we study *runtime costs* without treating a numerically invalid fast method as a winner.

The exact endpoints are interpolation at $s=0$ and the polynomial-constrained limit at $s=1$, which is evaluated analytically rather than factorizing a matrix at $\lambda=\infty$. Figure 1 illustrates why a user benefits from a common unit-interval smoothness coordinate even though the corresponding penalty scales vary with d.

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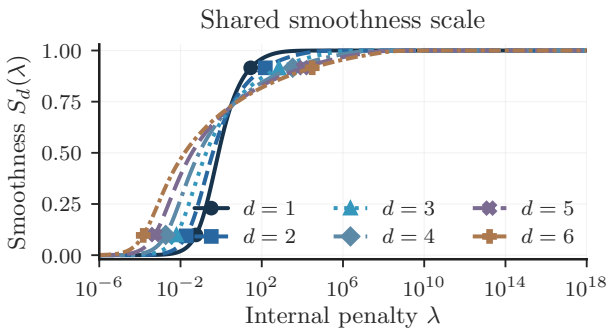


Figure 1: Smoothness versus penalty for $n = 48$, orders $d = 1, \ldots, 6$. The sixth-order line illustrates the mathematical mapping, not a completed Windows timing benchmark for $d = 6$.

1.2. Polynomial limits and endpoint conventions

At $S = 1$ the exact constrained fit minimizes $\|y - \tau\|^2$ subject to $D_d\tau = \mu\mathbf{1}$. For $\mu = 0$ its degree is at most $d - 1$; for fixed nonzero μ it may have degree d. This follows from the polynomial nullspace of the forward-difference operator and does not require factoring a matrix at $\lambda = \infty$. At $S = 0$, $\widehat{\tau} = y$ and the smoother is the identity; some cross-validation and information-criterion expressions are undefined at this interpolation endpoint. The penalties corresponding to the same interior S can differ considerably across d, which motivates using the normalized coordinate rather than a common raw penalty value.

1.3. Research questions

A single fitted curve, one criterion score, a complete M-point grid search, and R independent series define distinct workloads. We separate spectral or banded preparation, parameter inversion, fitted-curve reconstruction, trace/leverage computation, marginal likelihood determinants and repeated-series reuse. The theoretical question is not whether the bounded coordinate always saves time, but whether the *same exact candidate set* $\{\Lambda_{n,d}(s_j)\}$ can be shared across criteria and observed series without changing the statistical problem. Sections 2–4.6 establish the corresponding arithmetic and memory bounds.

2. Algorithmic time and storage complexity

Let n be series length, d difference order, $r = n - d$, K penalty candidates, R observed series, C scoring criteria and I root-solver evaluations per requested s. These are formal workload parameters; Section 4 derives detailed bounds for a set of criteria. All big-O statements are arithmetic models, not measured wall-clock rates. The penalty matrix $Q_d = D_d^\top D_d$ has half-bandwidth d.

2.1. Factorization and one complete fitted curve

Full SVD of the $r \times n$ difference matrix requires $O(nr^2)$ time and $O(nr)$ storage, or $O(n^3)$ and $O(n^2)$ respectively when d is small relative to n. Dense symmetric eigendecomposition of Q_d also has cubic time and quadratic storage; forming Q_d can exacerbate spectral conditioning. After spectral setup a full fitted curve uses $O(nr)$ products. Dense general solves and dense Cholesky require $O(n^3)$ per new penalty. General banded solving and banded Cholesky can factor in $O(nd^2)$ and use $O(nd)$ storage, followed by $O(nd)$ triangular or back substitution. Thus structured methods can be substantially faster for a *single fit*; Cholesky and general banded LU must not be conflated.

2.2. Criterion evaluation is not the same as fitting

For a fixed λ, the conditional effective degrees of freedom are $\nu(\lambda) = d + \sum_j (1 + \lambda\delta_j)^{-1}$ and normalized smoothness is (2), both computed in $O(r)$ after spectral preparation. A projected observation $z = V^\top(y - p_\mu)$ costs $O(nr)$ once, after which $\mathrm{RSS}(\lambda) = \sum_j w_j(\lambda)^2 z_j^2$ and the conditional GCV, AICc, BIC and profiled Gaussian marginal likelihood require $O(r)$ spectral sums per candidate. Classical fixed-μ LOOCV also requires leverages $(H_\lambda)_{ii}$ and residuals: calculating both using stored right singular vectors costs $O(nr)$ per penalty. This is *conditional observation-weight deletion*, not rolling-origin forecast validation, and empirical plug-in $\widehat{\mu} = \mathrm{mean}(D_d y)$ would not have the same exact leave-one-out interpretation without refitting that mean.

Nonspectral methods may fit quickly while incurring additional exact EDF and leverage work. Simply solving n banded right-hand sides to obtain inverse diagonals costs up to $O(n^2 d)$ beyond the $O(nd^2)$ factorization; specialized selected-inverse or structured algorithms can improve this (Weinert, 2007; Biessy, 2026). A fit-only cost must never be advertised as an end-to-end CV-selection cost.

For dense direct methods, EDF and LOOCV may be computed using a factorization and many right-hand sides; the displayed cubic bound reflects this full operation. A properly implemented banded selected inverse is not necessarily governed by the straightforward many-right-hand-side cost. These costs exclude software-level allocation and numerical conditioning, both measured or audited separately.

2.3. How pretranslation and grid reuse change total work

With the singular values cached, one inverse $s \mapsto \lambda$ is a monotone scalar root problem in $\log\lambda$ whose function calls each take $O(r)$, so K targets with I iterations cost $O(KIr)$ beyond the SVD. Let the set $G_{n,d} = \{\Lambda_{n,d}(s_j) : 1 \le j \le K\}$ be cached. It depends

Table 1
Dominant arithmetic for representative algorithm families. Costs are conditional on fixed μ and known finite λ. EDF and CV entries for nonspectral methods are not interchangeable with their fit cost. We state $r = n - d$ and assume a dense spectral basis is retained.

Method	Decomposition/setup	One fit	EDF/GCV	LOOCV	Memory
Fresh dense solve	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
Cached dense solve	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
Dense Cholesky	$O(nd^2)$	$O(n^3)$	$O(n^3)$	$O(n^3)$	$O(n^2)$
General banded LU	$O(nd^2)$	$O(nd^2)$	method-specific	method-specific	$O(nd)$
Banded Cholesky	$O(nd^2)$	$O(nd^2)$	method-specific	method-specific	$O(nd)$
Dense eigendecomposition	$O(n^3)$	$O(n^2)$	$O(n)$	$O(n^2)$	$O(n^2)$
Full SVD of D_d	$O(nr^2)$	$O(nr)$	$O(r)$	$O(nr)$	$O(nr)$

neither on the current series nor on the fixed mean of differences. The same grid can be reused across C criteria and R new series.

For GCV/AICc/BIC/ML-style spectral selections, a preliminary arithmetic bound is

$$T_{\mathrm{SVD}} = O(nr^2 + KIr + Rnr + RCKr). \tag{4}$$

The Rnr term projects each new series once; CV replaces the pertinent $O(r)$ candidate term by $O(nr)$. A proof with explicit memory and criterion-set assumptions appears in Section 4. A cold approach that re-inverts s at every criterion or re-decomposes for each series incurs additional repeated work. Likewise, a banded Cholesky implementation that caches the same K factors across R series may incur $O(Knd^2 + RKnd)$ fitting arithmetic, *before* trace/leverage computation. This is why different setup/reuse policies can reverse a timing winner.

The reference-mean policy changes coefficients of the right-hand side, not the spectrum. Fixed $\mu = 0$ and fixed $\mu = 0.15$ share precisely the same $G_{n,d}$. Estimating $\widehat{\mu}$ from $D_d y$ adds an $O(nd)$ pass per series; the archived comprehensive Windows run did *not* separately time this data-dependent mean-estimation step.

2.4. What is actually maximized under the Gaussian ML criterion?

The label *ML* means profiled Gaussian *marginal* variance-component likelihood, not pointwise maximum likelihood of the fitted curve. It assumes independent observational errors and a Gaussian penalty prior,

$$\begin{aligned} y &= \tau + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2 I), \\ D_d\tau - \mu\mathbf{1} &\sim N(0, \sigma_\eta^2 I), \quad \lambda = \sigma_\varepsilon^2/\sigma_\eta^2. \end{aligned} \tag{5}$$

The polynomial nullspace is unpenalized and eliminated from the reduced likelihood. On the $r = n - d$ penalized singular coordinates z_j, the marginal variances are $\sigma_\varepsilon^2 c_j(\lambda)$ with $c_j = 1 + 1/(\lambda\delta_j)$. Profiling the unknown observation variance gives

$$\begin{aligned} \widehat{\sigma}_\varepsilon^2(\lambda) &= \frac{1}{r}\sum_{j=1}^{r} z_j^2/c_j(\lambda), \\ -2\ell_{\mathrm{prof}}(\lambda) &= r\log\widehat{\sigma}_\varepsilon^2(\lambda) + \sum_{j=1}^{r}\log c_j(\lambda) + \text{constant}. \end{aligned} \tag{6}$$

This is a REML-like reduced-coordinate likelihood and it *does* depend on a variance model. After selection, one may also compute $\widehat{\sigma}_\eta^2 = \widehat{\sigma}_\varepsilon^2/\widehat{\lambda}$. The theoretical formulas condition on a fixed numeric μ; any data-derived $\mu = \mathrm{mean}(D_d y)$ would instead constitute a conditional plug-in, *not* joint maximum likelihood over $(\mu, \lambda, \sigma_\varepsilon^2, \sigma_\eta^2)$. For fixed variances and λ, maximizing the Gaussian posterior in τ recovers the ordinary PLS trend; that is a *different* optimization from selecting λ by marginal ML.

2.5. Matched versus unmatched coordinate searches

We separate *native s grid* (invert each s_j during a search), *premapped s grid* (invert once, pass the identical λ_j to the score), and an *independently spaced log-λ grid*. Only the first two evaluate the same statistical candidates, and hence they must agree in exact arithmetic. The third may run faster but selects among *different interior penalties*; its search time alone cannot establish an efficiency advantage at equal statistical quality. Continuous bounded optimizers in s and $\log\lambda$ follow different paths and are local searches, not certified global optima. Exact $s = 0$ and $s = 1$ endpoints are scored analytically.

The classical residual-based BIC expression has a divergent interpolation boundary; our numerical companion studies this limitation. The unrestricted residual-BIC expression is not statistically well-posed at this interpolation boundary.

3. Structured end-to-end criterion computation

A fast linear-system solution is not automatically a fast smoothing-parameter selector. For each candidate penalty, an implementation must obtain the trend, its residual sum of squares and, depending on the objective, the effective degrees of freedom (EDF), the diagonal leverage values or the marginal likelihood determinant. These auxiliary calculations are included in our definition of a *complete computational pipeline*. We give a structured construction that avoids a dense $n \times n$ inverse.

3.1. Banded factorization and the selected inverse

Let $A_\lambda = I + \lambda Q_d$, $Q_d = D_d^\top D_d$, and $\lambda \in (0, \infty)$. The symmetric positive definite matrix A_λ has half-bandwidth d. Factor it as $A_\lambda = L_\lambda L_\lambda^\top$ with lower triangular L_λ, whose nonzero entries satisfy $0 \le k - i \le d$. Define $Z_\lambda = A_\lambda^{-1}$. Although Z_λ is generally dense, only the entries Z_{ij} with $|i - j| \le d$ are needed in the selected-inverse recurrence. Backward substitution from $i = n$ to 1 gives

$$Z_{ij} = -\frac{1}{L_{ii}} \sum_{k=i+1}^{\min(n,i+d)} L_{ki} Z_{kj}, \quad i < j \le \min(n, i+d), \tag{7}$$

and

$$Z_{ii} = \frac{1}{L_{ii}^2} - \frac{1}{L_{ii}} \sum_{k=i+1}^{\min(n,i+d)} L_{ki} Z_{ki}. \tag{8}$$

Symmetry supplies Z_{kj} when $k > j$. These are selected-inverse (Takahashi-type) identities (Erisman and Tinney, 1975), not a proposal to store the full inverse. A banded factorization and selected inversion each require $O(nd^2)$ arithmetic, with $O(nd)$ factor and selected-entry storage for a fixed small difference order. The leverage diagonal is $h_i = Z_{ii}$ and $\nu(\lambda) = \sum_i h_i$.

The finite-penalty right-hand side is constructed directly:

$$b_{\lambda,\mu} = y + \lambda \mu D_d^\top \mathbf{1}, \qquad A_\lambda \widehat{\tau} = b_{\lambda,\mu}. \tag{9}$$

This bypasses the potentially huge polynomial particular solution p_μ, avoiding one source of catastrophic cancellation at large d and nonzero μ. It does not, however, guarantee forward accuracy when A_λ becomes badly conditioned. We require finite outputs, residual checks and independent reference comparisons before accepting a timed result.

3.2. Conditional criteria and computational requirements

For a *fixed numeric* μ define $e = y - \widehat{\tau}$, $\mathrm{RSS} = e^\top e$ and $h_i = (H_\lambda)_{ii}$, with $H_\lambda = A_\lambda^{-1}$. The classical leave-one-observation-out criterion and generalized cross-validation are

$$\begin{aligned} \mathrm{CV}(\lambda) &= \frac{1}{n} \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_i} \right)^2, \\ \mathrm{GCV}(\lambda) &= \frac{\mathrm{RSS}/n}{(1 - \nu(\lambda)/n)^2}. \end{aligned} \tag{10}$$

The residual-based information-criterion convention considered here has the form

$$\begin{aligned} \mathrm{BIC}_{\mathrm{RSS}}(\lambda) &= \log(\mathrm{RSS}/n) + \nu(\lambda) \log n / n, \\ \mathrm{AICc}(\lambda) &= \log(\mathrm{RSS}/n) + \frac{2\nu(\lambda) + 1}{n - \nu(\lambda) - 2}. \end{aligned} \tag{11}$$

We retain the exact score conventions of the existing software rather than silently substituting a different information criterion. The BIC expression decreases without bound as $\lambda \downarrow 0$ under a nondegenerate interpolation limit; the finite grid lower boundary must therefore be disclosed. The profiled Gaussian marginal-likelihood criterion additionally requires a weighted quadratic form and a log determinant on the penalized subspace, described in Section 2. Its implementation is not interchangeable with merely calculating a residual MSE.

If μ is estimated from the same observations as $\widehat{\mu} = \mathrm{mean}(D_d y)$, then H_λ is no longer the complete response-to-fit derivative of the plug-in procedure. The standard fixed-μ leverage formula in (10) is *not* an exact leave-one-out refit of this estimated mean. The fixed-mean theorems apply for any specified numeric μ, including zero and nonzero values; the plug-in case must be labeled as conditional.

3.3. Sparse and iterative methods with honest diagnostic costs

The implementation archive includes multiple end-to-end *pipelines*, not interchangeable Cholesky routines. Sparse SuperLU, general banded LU, SuiteSparse CHOLMOD, and preconditioned conjugate-gradient (PCG) each provide a fitted trend through a different algorithm. In the existing research code their CV/GCV/AICc/BIC/ML scores are completed by an *additional* banded selected-inverse diagnostic step. Thus any realistic hybrid cost includes both the named fit backend and the added diagnostic factorization. SciPy sparse LU is not CHOLMOD; an unavailable optional CHOLMOD installation is recorded as such.

PCG uses a matrix-free application of $I + \lambda D_d^\top D_d$ and FFT-periodic, DST-Dirichlet or Jacobi preconditioning. The FFT and DST symbols approximate the natural finite-series boundary structure; they are not exact diagonalizations, and no fixed small iteration

Table 2
Three exact, complete pipelines at a given finite penalty. The candidate penalty is already known; any cost of mapping $S \mapsto \lambda$ is accounted for separately. Here $r = n - d$ and the reported orders count arithmetic, not seconds. The banded entry includes the *entire* Cholesky and selected-inverse diagnostic, not just a fitted curve.

Pipeline	Reusable setup	Work for one candidate	Storage
Dense inverse	None	$O(n^3)$ inverse; $O(n^2)$ diagnostics	$O(n^2)$
Full SVD	$O(nr^2)$ once	$O(r)$ for a projected scalar score; $O(nr)$ when CV requires fitted residuals and leverage	$O(nr)$
Banded Cholesky with selected inverse	$O(nd^2)$ structural preparation	$O(nd^2)$ factor and diagnostics; $O(nd)$ for each fitted series	$O(nd)$

Table 3
Additional implementation-specific hybrid pipelines. These rows are *not* alternative asymptotic bounds for a pure factorization: each backend fit is supplemented by a separately charged $O(nd^2)$ banded diagnostic. The iteration count q of PCG is not assumed bounded independently of conditioning.

Hybrid pipeline	Fit phase	Complete-score overhead
Banded/sparse LU + trace	LU-dependent factorization and solve	Added $O(nd^2)$ banded selected inverse
CHOLMOD + trace	Sparse Cholesky factorization and solve	Added $O(nd^2)$ banded selected inverse
PCG + trace	$O(qnd)$ matrix-free iterations; transform preconditioners can add $O(qn \log n)$	Added $O(nd^2)$ banded selected inverse

count is guaranteed. Nonconverged PCG runs cannot enter the successful-time comparison.

For a single candidate, factorization cost and complete selection cost need not have the same ranking. Table 2 is used to derive the workload comparison in Section 5; Table 3 documents the implementation families without conflating their additional diagnostics with the named fitting backend.

4. An amortized cost model for complete smoothing-criterion workflows

We now distinguish a single mathematical evaluation from a *workload*. The former concerns the arithmetic complexity of solving a PLS problem at one fixed penalty; the latter concerns candidate searches over multiple independent series and multiple selection criteria. The distinction is crucial because the smoother matrix and the smoothness-to-penalty map are independent of the observed series for fixed (n, d).

4.1. Workload definition and numerical admissibility

Fix sample length n, difference order $1 \le d < n$, and write $r = n - d$. Let $\mathcal{S} = \{s_1, \dots, s_M\} \subset (0, 1)$ be a specified set of M *distinct* interior normalized smoothness levels, and let $y^{(1)}, \dots, y^{(R)} \in \mathbb{R}^n$ be R observed series. A set $\mathcal{C}$ of statistical criteria is evaluated on identical penalties $\lambda_j = \Lambda_{n,d}(s_j)$. The criterion label C is allowed to be a singleton (e.g., CV or GCV) or a set of criteria, so $c = |\mathcal{C}|$ need not equal one. Define

$$T_{\mathcal{A}} = T_{\text{setup}} + T_{\text{map}} + T_{\text{factors}} + T_{\text{series}} + T_{\text{scores}} + T_{\text{overhead}}. \tag{12}$$

Here $\mathcal{A}$ denotes the complete numerical pipeline, ε its requested accuracy tolerances, and θ the implementation and machine constants. A value of $T_{\mathcal{A}}$ is an *execution time*; an asymptotic bound on its arithmetic work is not an observed number of milliseconds. We use a single-threaded exact-arithmetic cost model for proofs, with floating-point admissibility checked separately.

A pipeline is *admissible* on a prescribed workload only if every required score is defined, its numerical checks pass, and its working memory fits the available budget. A small runtime on a failed or truncated search does not define a valid winner. This constraint makes the theoretical frontier a conditional statement rather than a universal solver ranking.

4.2. Proposition: reuse of a smoothness grid

Proposition 1 (data-independent penalty grid). For fixed (n, d), the inverse image $\{\Lambda_{n,d}(s_j)\}_{j=1}^{M}$ is the same for every observed series, every fixed numerical difference mean, and every criterion in $\mathcal{C}$.

Proof. Equation (2) depends only on the nonzero eigenvalues of $D_d^\top D_d$. For each finite $\lambda \ge 0$, its derivative is

$$S_d'(\lambda) = \frac{1}{r} \sum_{j=1}^{r} \frac{\delta_j}{(1 + \lambda \delta_j)^2} > 0.$$

Thus each $s_j \in (0, 1)$ has a unique inverse λ_j that is independent of y, μ and the criterion score. Calculating

the inverse twice for the same (n, d, s_j) cannot change the mathematical candidate. □

The proposition does *not* imply that an independently spaced grid in $\log\lambda$ has the same candidates. It also does not imply that continuous local optimizers in s and $\log\lambda$ traverse the same numerical path.

4.3. Proposition: complete criteria from banded information

Proposition 2 (banded sufficiency). At fixed finite $\lambda > 0$ and fixed numeric μ, an SPD banded Cholesky factorization of $A_\lambda = I + \lambda D_d^\top D_d$, together with the selected inverse on its sparsity band, suffices to evaluate the fitted curve, conditional CV/GCV/AICc/residual BIC, and profiled Gaussian marginal likelihood, without forming a dense $n \times n$ inverse.

Proof. The direct normal equations are

$$A_\lambda \widehat{\tau}_\lambda = y + \lambda\mu D_d^\top \mathbf{1}. \tag{13}$$

One banded triangular solve gives $\widehat{\tau}$. A backward selected-inverse recursion (Section 3) gives $h_i = (A_\lambda^{-1})_{ii}$, and hence $\nu = \sum_i h_i$. Residuals $e = y - \widehat{\tau}$ and RSS $= e^\top e$ then determine CV, GCV, AICc and residual BIC through Eqs. (10)–(11).

For the Gaussian reduced-coordinate likelihood, choose p_μ with $D_d p_\mu = \mu\mathbf{1}$, and put $z = y - p_\mu$. Let $w_j = \lambda\delta_j/(1+\lambda\delta_j)$. Then

$$\begin{aligned} J_\lambda &:= \sum_{j=1}^{r} w_j (v_j^\top z)^2 \\ &= z^\top (I - H_\lambda) z \\ &= \|y - \widehat{\tau}_\lambda\|^2 \\ &\quad + \lambda \|D_d \widehat{\tau}_\lambda - \mu\mathbf{1}\|^2. \end{aligned} \tag{14}$$

The final equality follows by substituting the normal-equation minimizer in the strictly convex penalized objective. Thus J_λ is available from the fitted curve and a banded difference pass; there is no need to construct the potentially ill-scaled polynomial p_μ.

For $\lambda > 0$, the marginal variance multipliers are $c_j = 1 + 1/(\lambda\delta_j)$. Since the r nonzero eigenvalues of Q_d are the eigenvalues of $D_d D_d^\top$, the determinant identity is

$$\begin{aligned} \sum_{j=1}^{r} \log c_j &= \sum_{j=1}^{r} \log(1+\lambda\delta_j) - r\log\lambda - \sum_{j=1}^{r} \log\delta_j \\ &= \log\det A_\lambda - r\log\lambda - \log\det(D_d D_d^\top). \end{aligned} \tag{15}$$

The first log determinant is the sum of twice the logarithms of the diagonal Cholesky entries of A_λ. The last determinant is independent of λ, y and μ and can be prepared once using a banded factorization of $D_d D_d^\top$. Profiling the observational variance gives $\widehat{\sigma}_\varepsilon^2 = J_\lambda/r$ and a negative twice-log-likelihood, up to constants,

$$\begin{aligned} \mathcal{L}(\lambda) &= r\log(J_\lambda/r) + \log\det A_\lambda \\ &\quad - r\log\lambda - \log\det(D_d D_d^\top). \end{aligned} \tag{16}$$

Consequently all the stated quantities can be evaluated from banded information alone. □

The likelihood statement assumes the latent-difference Gaussian variance model of Eq. (5). It is not ordinary residual least squares and is not joint likelihood optimization over μ. For $\mu = \mathrm{mean}(D_d y)$ computed from the same series, the fixed-μ leverage formula is *conditional*, not an exact leave-one-out refit of the plug-in estimator. At $\lambda = 0$ some selection formulas are undefined and the Gaussian prior limit is improper; $s = 0$ and $s = 1$ must be treated by their own endpoint conventions.

4.4. Theorem: arithmetic and memory bounds

We assume a conventional full retained SVD of D_d, an SPD banded factor and Takahashi-type selected inversion, and fixed d-banded operations. Set $\chi_{\mathrm{CV}} = 1$ when ordinary conditional CV belongs to $\mathcal{C}$ and zero otherwise. Let I be a uniform upper bound on the number of function evaluations used to invert each interior s_j to the required scalar tolerance.

Theorem 1 (amortized work). With reusable full spectral preprocessing and a single shared smoothness-to-penalty map, the arithmetic work for R series and M candidates is bounded by

$$W_{\mathrm{SVD}} = O\big(nr^2 + MIr + Rnr + MRcr + \chi_{\mathrm{CV}} MRnr\big). \tag{17}$$

Its retained dense-basis memory is $O(nr)$, in addition to $O(M + Rn)$ if all the observed series are held simultaneously.

Using banded Cholesky and exact selected-inverse diagnostics, and evaluating $S_d(\lambda)$ by banded trace calculations during root finding, the arithmetic work is bounded by

$$W_{\mathrm{band}} = O\big(nd^2 + MI\,nd^2 + Mnd^2 + MRnd + MRcn\big). \tag{18}$$

When the penalty grid has already been mapped, the $MI\,nd^2$ term is absent. A streamed implementation retains $O(nd+M+Rn)$ storage; retaining *all* M banded factors instead requires $O(Mnd)$ storage.

Proof. The reduced SVD has cost $O(nr^2)$ and stores $O(nr)$ entries. Each spectral smoothness evaluation is a sum of r scalars, so M inversions of at most I evaluations cost $O(MIr)$. One projection per observed series requires $O(Rnr)$. Once projected, any of the c scalar scores at each (j,k) is $O(r)$. If CV is included, exact leverage and full residual reconstruction are bounded by $O(nr)$ per candidate and series, giving the additional CV term. This is a conservative bound; leverage vectors can be shared among the R series at each candidate.

For the banded method, preparation of the banded difference-penalty matrix and the constant Gaussian

determinant costs $O(nd^2)$. Every trace-based root-solver evaluation factors an SPD matrix and computes the needed selected inverse in $O(nd^2)$ operations. After mapping, factorization and selected inversion are done once per candidate in $O(Mnd^2)$, reused across all criteria and all observed series. For each candidate and series, two triangular solves, difference passes and quadratic summaries use $O(nd)$ arithmetic, followed by at most $O(cn)$ score formation. Processing one candidate at a time requires one banded factor and selected inverse; keeping all observations uses $O(Rn)$ additional storage. Caching all factors produces the stated $O(Mnd)$ alternative. □

These are upper bounds for explicit, fully specified implementation families. They do not prove a lower bound for every other possible algorithm; fast transforms for selected difference orders or alternative numerical approximations can produce different costs. Furthermore, a large root-iteration count I can erase the benefit of banded factorization when the user insists on inverting an extremely fine s grid.

4.5. Corollary: order-one difference spectra

Corollary 1. For the first-difference matrix with the ordinary finite-interval boundary convention,

$$\delta_j = 2 - 2\cos(\pi j/n), \qquad j = 1, \dots, n-1. \quad (19)$$

Thus the entire positive penalty spectrum is available in $O(n)$ arithmetic and storage without a dense SVD, and the M-point smoothness grid can be inverted in $O(MIn)$ arithmetic through direct spectral sums.

Proof. $D_1^\top D_1$ is the path-graph Laplacian with endpoint diagonal values one, interior diagonal values two, and adjacent off-diagonals -1. Its cosine eigenvectors have eigenvalues $2 - 2\cos(\pi j/n)$, $j = 0, \dots, n-1$. The zero eigenvalue corresponds to the constant vector. Summing the remaining $n-1$ eigenvalues in Eq. (2) takes $O(n)$ arithmetic per root evaluation. □

This special case is an established discrete-cosine identity, not a claimed new spectral algorithm. For general $d > 1$, endpoint effects mean that the same elementary eigenvalue expression cannot simply be substituted.

4.6. When can a crossover exist?

A method crossover depends on the machine, statistical workload, numerical tolerance and actual pipeline implementation. To expose this dependence, write a local empirical calibration model for a *fixed* $(n, d, R, \mathcal{C}, \varepsilon, \theta)$:

$$T_{\mathcal{A}}(M) = a_{\mathcal{A}} + b_{\mathcal{A}} M + \rho_{\mathcal{A}}(M), \quad (20)$$

where $a_{\mathcal{A}} \geq 0$ is fixed preparation, $b_{\mathcal{A}} \geq 0$ is per-candidate effective work at that workload, and $\rho_{\mathcal{A}}(M)$ records cache changes, allocation, nonlinear solver iterations and contention effects. This is a *calibration ansatz*, not an exact equality implied by big-O.

Proposition 3 (conditional crossover). If $\rho_{\mathrm{SVD}} = \rho_{\mathrm{band}} = 0$ over a declared workload interval, $a_{\mathrm{SVD}} > a_{\mathrm{band}}$ and $b_{\mathrm{band}} > b_{\mathrm{SVD}}$, then banded Cholesky is faster for $M < M_\star$, spectral SVD is faster for $M > M_\star$, and

$$M_\star = \frac{a_{\mathrm{SVD}} - a_{\mathrm{band}}}{b_{\mathrm{band}} - b_{\mathrm{SVD}}}. \quad (21)$$

If the sign conditions do not hold, no positive crossover of that direction follows from the affine model.

Proof. Subtract the two affine time functions:

$$T_{\mathrm{SVD}}(M) - T_{\mathrm{band}}(M) = (a_{\mathrm{SVD}} - a_{\mathrm{band}}) - M(b_{\mathrm{band}} - b_{\mathrm{SVD}}).$$

Its unique zero is $M_\star > 0$ under the stated inequalities, and its sign changes once at this point. □

The proposition does *not* supply a universal observed threshold: constants must be calibrated on comparable, numerically admissible workloads. When ordinary CV is required, the $O(nr)$ spectral candidate work may change the signs. When multiple series share an already-mapped s grid and reuse banded factors, the effective b_{band} can decrease. Consequently, a universal difference-order-only dispatcher is not justified by these complexity bounds.

4.7. Finite-memory frontier and candidate streaming

For an eight-byte floating-point type, one dense $n \times n$ matrix occupies $8n^2$ bytes before copies or solver workspaces. At $n = 131\,072$ this is 128 GiB, so an ordinary dense inverse can fail the memory budget before any statistical criterion is evaluated. Banded factorizations instead require $O(nd)$ storage for fixed d. An implementation may stream penalty candidates one at a time, updating the best valid criterion score per series and criterion, without retaining M full smoothing matrices.

The total-time functional (12) should therefore be minimized over the *admissible set* of algorithms:

$$\mathcal{A}^\star \in \arg\min_{\mathcal{A} \in \mathfrak{A}_{\mathrm{valid}}} T_{\mathcal{A}}(n, d, M, R, \mathcal{C}; \varepsilon, \theta). \quad (22)$$

Here $\mathfrak{A}_{\mathrm{valid}}$ is workload-specific: statistical undefinedness, forward-error limits, numerical nonconvergence and the memory budget can remove a method from the comparison. The formula is a decision framework and not a claim that its minimizer has been empirically identified across the full design space.

5. Practical algorithm regimes and worked illustrations

Theorems provide work and storage bounds, but a reader must also know what changes when the scientific workload changes. We distinguish a *derived preference under stated assumptions* from an *observed numerical comparison*. The former cannot be read off from milliseconds, and the latter cannot establish an asymptotic ordering.

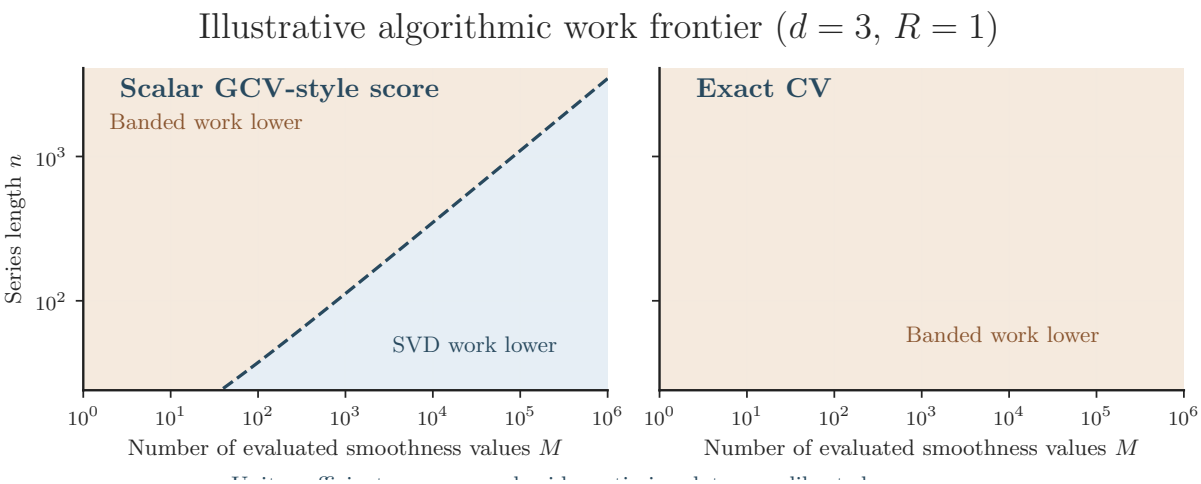


Figure 2: Conceptual two-criterion frontier derived from an explicit *unit-coefficient arithmetic proxy*, not benchmarked seconds. For $d = 3$, $R = 1$ and an already mapped penalty grid, blue denotes lower modeled full-SVD work and warm shading lower modeled banded-Cholesky work. The dashed curve is the proxy's equality locus, not a measured decision boundary. CV includes the dense leverage and fitted-residual cost; GCV uses scalar spectral scoring. No numerical conditioning or hardware-dependent constants are inferred from the shading.

5.1. When is SVD preferable, and when is Cholesky?

For a fixed (n, d), a full retained SVD requires $O(nr^2)$ preprocessing and $O(nr)$ memory ($r = n - d$). Subsequent projected scalar GCV, AICc, residual BIC and profiled Gaussian likelihood scores are $O(r)$ per candidate and series after projections. This favors spectral reuse *conditionally* when n is moderate, the number M of smoothness values is large, and the same decomposition serves many observed series or criteria.

A banded Cholesky factorization with an exact selected-inverse diagnostic instead requires $O(nd^2)$ arithmetic per penalty and $O(nd)$ working memory at fixed small d. It is particularly attractive for large n, small or moderately sized grids, and memory-constrained problems; the per-candidate diagnostic is charged even when only the fitted trend would otherwise be inexpensive. A generic sparse Cholesky backend can exploit the same structure, but a hybrid that performs an additional banded factorization should not be called a pure sparse-Cholesky timing.

For *conditional ordinary CV*, the spectral method needs the leverage diagonal $h_i = \sum_k v_{ik}^2/(1 + \lambda\delta_k)$ and the fitted residual vector, costing up to $O(nr)$ per new candidate; the leverage is reusable across the R series. By contrast, the banded selected-inverse recursion obtains all h_i with the same $O(nd^2)$ factorization order. Therefore the crossover for CV need not coincide with the scalar-GCV crossover. In the fixed-d, increasing-n asymptotic regime, the banded CV path has a structural arithmetic advantage over the full dense-spectral CV calculation. Nevertheless, optimized matrix kernels and numerical tolerances can reverse a *finite-size measured* ranking. Neither family is universally the fastest.

For a fixed problem size, Proposition 3 permits a *positive* $M_\star$ only when the SVD has a larger preparation cost but a strictly smaller marginal cost of each additional candidate. Figure 3 illustrates both the possible scalar-criterion crossover and the absence of one under its chosen CV proxy. Figure 2 similarly provides an interpretation of the work model, not a claimed universal machine-level speed classifier.

Selection policy. For a requested full grid, first decide which *complete* criterion is required. Scalar spectral scores and classical fixed-μ CV do not have the same candidate costs. Next apply the dense-memory budget before comparing nominal arithmetic bounds; a full basis cannot be presumed feasible for very large n. Finally, make the choice only between numerically admissible pipelines. An accurate selected-score match in float64 is useful evidence but is not a high-precision forward-error proof.

5.2. One reproducible implementation comparison

A narrowly scoped benchmark illustrates why the theoretical workload variables matter. We use the existing archived Windows 11 Experiment010 result for $n = 64$, $d = 1$,

$$\widehat{\mu} = \mathrm{mean}(D_1 y), \qquad \mathcal{S} = \{j/10^5 : 1 \le j \le 10^5\},$$

and the source implementation's residual-BIC score. The end of the grid is the exact $S = 1$ constrained limit; it is not an attempt to invert an infinite penalty. Each method executed two recorded repeats for each of the four one/multiple-candidate, one/multiple-series workloads and both *direct-S* and already mapped-λ coordinates. We report the matched mapped-λ observations below.

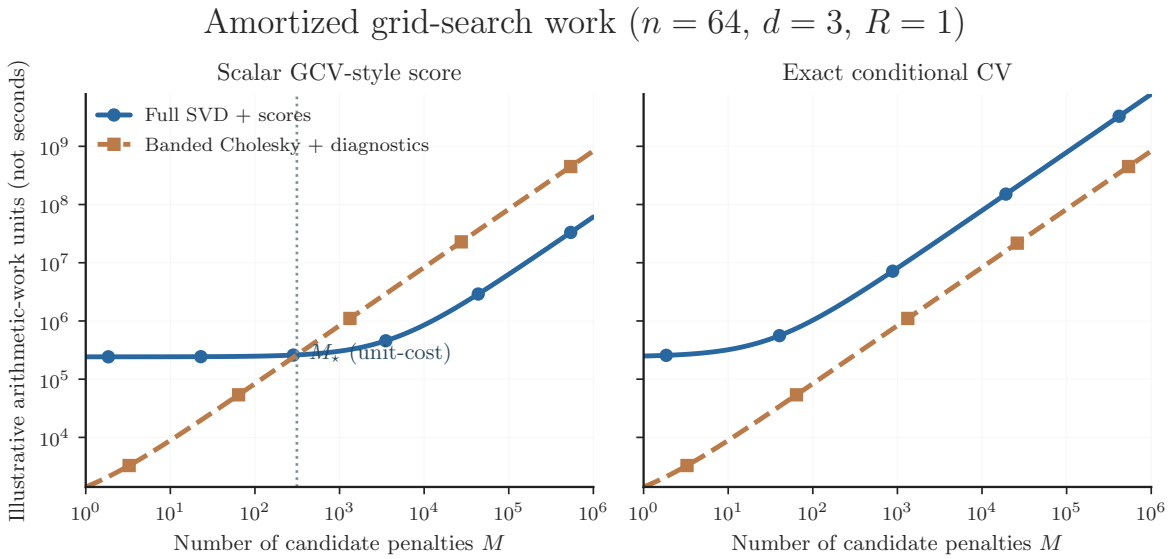


Figure 3: Amortized *operation-work proxies* as M grows, with $n = 64$, $d = 3$, $R = 1$ and identical premapped penalties. All leading terms are assigned unit coefficients for illustration; the marked $M_\star$ in the scalar-score panel is not an experimentally calibrated runtime crossover. The CV panel charges full spectral leverage/residual reconstruction and the banded selected inverse. Axes are dimensionless arithmetic counts, not milliseconds.

Table 4
Measured median total cold elapsed time from the *matched single-worker* Windows run. Single-point times are in milliseconds; the 10^5-candidate searches are in seconds. Each entry is the median of two repeated source records at fixed $n = 64$, $d = 1$ and BIC, with the penalty grid mapped in advance under the benchmark's stated cold-timing convention. Do not treat the differing units as a speedup.

	Single S (ms)		10^5 values (s)	
Pipeline	$R = 1$	$R = 2$	$R = 1$	$R = 2$
Full SVD	2.130	2.150	8.625	9.262
Dense inverse	1.932	1.955	11.603	15.384
CHOLMOD hybrid	2.606	2.676	32.003	37.603

All three methods were run sequentially with one worker and one BLAS thread on the same university Windows machine, under the same source revision, seed, candidate set and numerical criterion. The sparse method is a *CHOLMOD plus banded selected-inverse* hybrid, not a pure CHOLMOD factorization. All 48 raw outcome records have successful status and matching selected criterion scores to the recorded tolerance. The comparison is limited to one $(n, d, \text{criterion})$ configuration and two repeats; it does not estimate universal crossover thresholds.

The fully auditable CSV is frozen at `data/experiment01_matched_three_methods.csv` with original source SHA and concurrency labels. It is traceable to the timestamped results branch described in `THEORY_HANDOFF.md`. The one-worker comparison is an implementation *example*, not the primary mathematical contribution of this paper. In particular, the residual-BIC interpolation boundary may produce a boundary minimizer rather than a well-defined unrestricted optimum, so these runtime measurements must not be interpreted as evidence that BIC is statistically optimal for trend forecasting.

5.3. Real-data illustration: third differences as jerk

A separate application manuscript in this repository studies measured longitudinal vehicle positions from the FHWA NGSIM US-101 dataset (U.S. Department of Transportation, Federal Highway Administration, 2016). We reuse only one *predeclared illustrative* window from that study to demonstrate how a workload-defined matched three-method pair maps to a physical signal. The source contains 10-Hz camera-derived positions in meters, and the training window has $n = 80$ observations. The present illustration uses $d = 3$ and a training-only plug-in difference mean

$$\widehat{\mu} = \frac{1}{n-3}\sum_{t=1}^{n-3}(D_3 y)_t, \qquad \frac{D_3\tau}{(\Delta t)^3} \approx \tau^{(3)}(t).$$

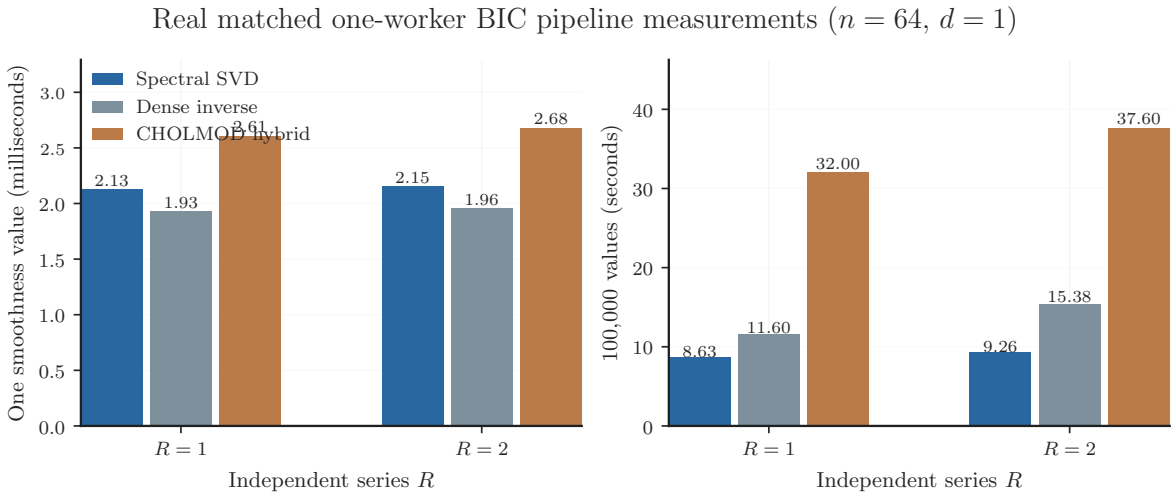


Figure 4: Recorded cold-time medians in Table 4, recomputed from the 48 original source rows. Left: one candidate in milliseconds. Right: 10^5 candidates in seconds. All values come from one-worker, single-BLAS-thread runs. They demonstrate the benefit of amortized SVD evaluation for this *particular* BIC configuration; CV and larger-n conclusions require their own matched numerical comparisons.

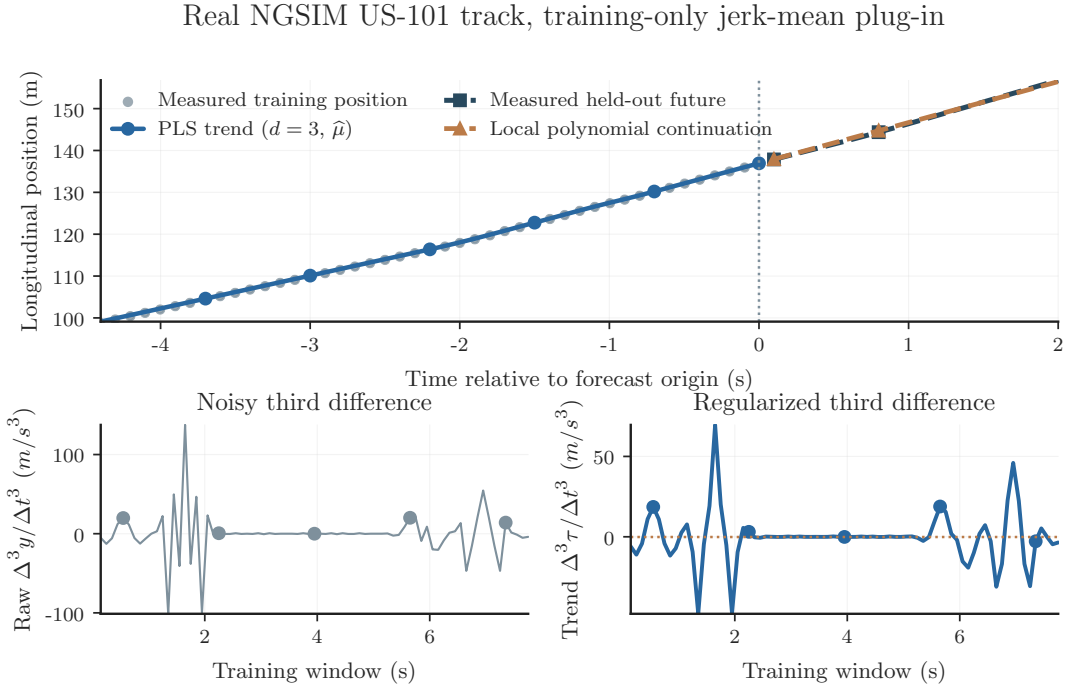


Figure 5: One real US-101 trajectory, reused from the separate NGSIM jerk case study. Upper panel: the last 45 measured training positions, $d = 3$ PLS fit with training-derived $\widehat{\mu}$, measured future positions held out during fitting, and a *separately appended* local-polynomial continuation over two seconds. Lower panels: noisy third differences of observations and of the regularized trend, on their individual vertical scales. These curves do not establish a true jerk value or a PLS-intrinsic forecast.

The latter is a *jerk proxy* for position sampled at interval $\Delta t = 0.1$ s, not ground-truth jerk from an accelerometer. The smoothing parameter is the previously selected GCV grid candidate $S = 0.40$, $\lambda \approx 0.08084$, without using any held-out positions during the selection. Because $\widehat{\mu}$ is data-dependent, its ordinary CV leverage formula would be conditional rather than an exact refitting of the plug-in mean.

PLS itself estimates the historical trend and does *not* produce the forecast in Figure 5. The companion study obtains future positions by fitting a local polynomial of degree $d-1$ to the last 25 fitted positions after removing the particular solution $\widehat{\mu}\binom{t}{d}$, then restoring that particular solution at the future indices. The hold-out comprises 20 chronological frames, and no model is fitted to those future observations. The full evaluation,

physical baselines and the distinct forecasting objectives belong to `paper/ngsim-jerk-case-study/`; the present paper uses only the illustration to connect its abstract cost variables to a concrete time-series workload. Numerical differentiation of noisy observations is itself an ill-posed estimation problem, as discussed by van Breugel, Kutz and Brunton (2020).

6. Implementation-independent computational policy

The preceding bounds suggest a reproducible decision procedure, but not a universal numerical winner. In practice the expensive operations must be attributed to the correct level of reuse: a factorization is specific to (n, d, λ), whereas a smoothing-grid inverse is specific only to (n, d, s). A fitted curve is specific to the observations and to the reference-difference mean. A scalar selection criterion depends on the already-computed residual and diagnostic summaries, whereas its numerical definition is unchanged by a reparameterization of the candidate grid.

6.1. Cost-aware organization of a complete search

For each fixed (n, d) the following abstract procedure avoids unnecessary repetitions. It is pseudocode for organizing computations, not a claim of a newly implemented universal dispatcher.

1. Fix the requested criterion set $\mathcal{C}$, the target-smoothness grid $\mathcal{S}$, numerical tolerances, and fixed or explicitly conditional plug-in means. Determine whether any requested criterion requires per-observation leverages (ordinary CV) or a Gaussian marginal determinant.
2. Choose a mathematically admissible smoothness mapping method. Invert each interior s_j *once*; handle $s = 0$ and $s = 1$ by their respective analytical limits. Record a bracketed inversion residual or a rigorous stopping condition, not merely an iteration count.
3. Choose the candidate-score family subject to the memory budget and required outputs. For the spectral family, prepare the basis once and the projected observations once per series. For the banded family, factor each A_{λ_j} once and obtain its selected inverse or other needed diagnostics once.
4. For each $j = 1, \ldots, M$, reuse the same penalty and factorization across every observed series and every criterion that permits such sharing. Calculate all required score terms, flag undefined scores and numerical failures, and update the best valid candidate for each series–criterion pair.
5. Record the decomposition, grid mapping, criterion evaluation, memory and accuracy costs *separately*; report the complete elapsed time only for workloads with all required results.

If no score requires the full fitted curve, computing it for every candidate is unnecessary for the spectral family. Conversely, ordinary conditional CV requires leverage diagonals and residuals, so a fit-only solver time is not a full CV-selection time. Marginal Gaussian likelihood also requires its own variance and determinant assumptions; the least-squares objective should not be substituted for the likelihood merely because both involve $D_d^\top D_d$.

6.2. Rival algorithmic explanations

One explanation for good observed performance is *arithmetic sparsity*: the d-banded positive definite system admits a direct solve and its selected-inverse diagnostic in nearly linear arithmetic for fixed d. This is the strong case for banded Cholesky when n is large.

A rival explanation is *amortization*: dense spectral preprocessing is expensive, but the same singular basis can serve many smoothed-series and scalar-criterion evaluations. The second explanation is especially relevant for large M and scalar GCV/AICc/BIC/ML scoring. Neither explanation dominates without the actual workload and implementation constants, and both can be affected by numerical conditioning. The synthesis is the conditional admissible frontier of Eq. (22), rather than a globally preferred factorization.

The importance of exploiting Whittaker–Henderson structure was demonstrated long before this paper: Weinert (2007) developed efficient fitted trends and GCV computations, and Biessy (2026) examined banded methods and reduced-rank parameterizations in a modern statistical framework. The sparse selected-inverse principle is likewise classical (Erisman and Tinney, 1975). Our contribution is an explicit *combined workload model* that keeps normalized-smoothness inversion, criterion diagnostics, reuse, endpoints, numerical admissibility and memory on the same accounting basis. It is not a claim to have invented Cholesky, selected inversion, the Gaussian variance model or the original smoothness index.

6.3. Limits of the theoretical results

Big-O bounds omit constants, BLAS scheduling, cache locality, dense-SVD numerical tolerances, Python dispatch, allocation, garbage collection and CPU contention. The threshold in Eq. (21) can be fitted only from controlled measurements; using a few timings collected under different concurrency conditions would be misleading. Furthermore, the numerical advantage of shift-free spectral reconstruction at large d or extreme λ is a separate accuracy concern, not an asymptotic runtime guarantee.

The statements concern a finite, equally spaced univariate series and a conventional forward difference penalty. Irregularly spaced observations, missing-data weights, nonquadratic losses, spatial lattices and higher-dimensional penalties can change the sparsity pattern and invalidate a naive reuse of $O(nd^2)$. Although the fitted trend is commonly described as a time-series trend, no in-sample conditional CV expression presented here establishes out-of-sample forecasting accuracy.

The endpoints also require care: $S(0) = 0$ returns the observations, but CV and information criteria can be undefined or divergent at exact interpolation; $S = 1$ means the constrained polynomial limit, not the inversion of A_λ at an artificially enormous finite penalty. A user-provided μ is fixed when the derivatives and conditional EDF are computed. If μ is estimated from y, this data dependence cannot simply be suppressed when deriving the exact leave-one-out score.

7. Conclusions

Smoothness-indexed finite-difference PLS admits two complementary computational representations. Its reusable spectrum supplies a monotone and data-independent map between normalized smoothness and the penalty; its banded normal equations supply a memory-efficient way to obtain fitted curves and exact conditional criterion diagnostics. We derived a complete banded expression for the reduced-coordinate Gaussian determinant and profiled quadratic form, established amortized work and storage bounds for repeated candidates, criteria and series, and expressed the implementation decision as an admissible, workload-specific cost frontier.

These deductions do not determine a universal fastest method in milliseconds. Section 5.2 reports one matched, single-worker BIC implementation example; the observed spectral advantage for a 10^5-candidate grid is descriptive at $n = 64$ and $d = 1$, not a claim about CV or arbitrary sample sizes. The NGSIM application in Section 5.3 likewise demonstrates the meaning of the estimated third difference and a separately appended forecasting heuristic, not an intrinsic PLS predictive distribution or measured true jerk. A fully powered multi-criterion runtime frontier and independent physical forecasting evaluation remain appropriate subjects for separate papers.

Data and code availability

The full mathematical derivations and reproducibility notes are maintained at https://github.com/heri-espino/trendestimation. An archived source revision must be cited at submission. The mathematical identities require no experimental calibration. The small matched benchmark illustration uses 48 frozen Windows observations and their original source revision, copied verbatim under `paper/csda-computational-costs/da`. The jerk example reuses real FHWA US-101 longitudinal measurements and the predeclared training-only selection from `paper/ngsim-jerk-case-study/`; the recorded positions and CC BY-SA 3.0 attribution are retained in that source manuscript. The four new AwesomeFigures sources, editable PGF/PDF/SVG outputs, figure-generation script, checksums, and full archived benchmark provenance are preserved in this repository.

Declarations

Author order, correspondence, funding, potential conflicts and contributor statements remain subject to confirmation by both authors before journal submission.

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